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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 306: Machine Learning Lab
Practice Assignment 3

Instructions: This is only for practice. Complete it by 12:00 PM today. Your completion will be reviewed by the Teaching Assistants.

1. Download [The salary dataset](#) regarding the prediction of salary from the year of experience. Write a program to do the followings:

- (a) Read the dataset.

- (b) Randomly split the dataset in percentage of training samples (NTs): Percentage of test samples ($NTes$) ratio .

Here, the values of NTs and $NTes$ are as follows ($NTs + NTes = 100\%$ always):

$NTs = 10:10:90$ (i.e., initial: 10%, increment by 10%, maximum is 90%)

$NTes = 90:10:10$ (i.e., initial: 90%, decrement by 10%, minimum is 10%)

- (c) Design different hypothesis ($\hat{y}$) to predict the salary from the year of experience using Gradient Descent (GD) method, **without using in-built Python packages /libraries** for estimation of the parameters, considering the generated training-testing splits from 1(b). Consider theta (θ) is initialized to zero and the learning rate (α) is set to 0.001. Separately save the model parameters for all the training splits.

- (d) Report the followings in form of a folder for plots/graphs (store directly from code), excel sheet/ CSV files for results (store directly from code), and the word document for result analysis/ interpretation:

- i. Store the plots (in the designated folder directly from code) of different estimated lines corresponding to different hypotheses, obtained in 1(c) over the various training splits.

- ii. Calculate and store (in the designated excel sheet for results directly from code) the prediction of salary for the test samples over different test splits considering the respective hypothesis.

- iii. Calculate and store (in the designated excel sheet for results directly from code) the coefficient of determination (R^2) and mean of sum of squared residuals (mean-RSS) for different training-testing splits separately.

- iv. Plot the obtained results of mean-RSS values in a graph between percentage of training samples in X-axis and values of mean-RSS for respective training and testing sets in Y-axis.

- v. Plot the obtained results of R^2 scores in a graph between percentage of training samples in X-axis and values of R^2 for respective training and testing sets in Y-axis.

- vi. Select the best training-testing split (in terms of (R^2) or mean-RSS). Using the selected split, calculate the model parameters and design hypotheses, considering various θ -initializations as zero, random values in the range $[0,1]$ and $[0,100]$. Store the obtained model parameters, (R^2) and mean-RSS scores for all the three cases.

vii. Using the selected best training-testing split, calculate the model parameters and design hypotheses, considering various θ -initializations as zero, random values in the range $[0,1]$ and $[0,100]$, and learning rates (α) as 0.0001, 0.05, 0.1, 1, 10, 100, and 1000. Store the obtained model parameters, (R^2) and mean-RSS scores for all the possible combinations of θ -initializations and learning rates.

viii. Write your own results analysis in the word document from the results and plots/graphs.

2. Execute the above assignment (in Q1) using in-built package/library for Gradient Descent.
3. Compare the various results obtained in Q1 (from i-v), with the ones obtained using OLS and interpret your analysis.